

The normalized Erdős–Straus quartic and its exact passage through the signed Fable cover to a fixed weighted conductor

The Clankers

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Abstract

An ordered positive solution of the Erdős–Straus equation determines a normalized binary quartic whose four literal roots are the prime and its three denominators. We prove that this coefficient target lies in the genuine degree-eight finite étale domain of the four-dimensional Fable map. The prime is intrinsic in the unmarked coefficients, an explicit hypersurface equation characterizes the algebraic image on the relevant open chart, and the two inverse states marking the prime are given in every source coordinate. For the exterior states developed in this workbench all four roots are distinct; sorting recovers the denominator marking and the complete Turn 7 coordinates. We also retain, as a different map with a different fibre, the earlier reciprocal-cubic suspension: it lies on the nonproper hyperplane and has seven states. Finally, explicit invertible linear maps carry both targets, their fibres and their exceptional loci into one fixed Meixner–Pollaczek weighted-conductor four-plane. These are exact arithmetic–geometric–operator bridges. They neither prove universal Erdős–Straus existence nor identify the four arithmetic roots with zeros of the Riemann zeta function.

1 The arithmetic source and the four-dimensional map

Retain an ordered positive integral solution

$$\frac{4}{p} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z}, \quad p, x, y, z \in \mathbb{Z}_{>0}. \quad (\text{EZ1})$$

Put

$$s_1 = x + y + z, \quad s_2 = xy + xz + yz, \quad s_3 = xyz, \quad S = p + s_1. \quad (\text{EZ2})$$

Multiplication of EZ1 by $pxyz$, with no division by a possibly vanishing coordinate, gives the exact relation

$$ps_2 = 4s_3. \quad (\text{EZ3})$$

In source coordinates $(a, y_0, z_0, w_0) \in \mathbb{C}^4$, let

$$\begin{aligned} P_0 &= a^3 z_0 + 2a^2 y_0 - ia, \\ P_2 &= -a^3 y_0^2 z_0 - 2ia^2 y_0 z_0 + a^2 w_0 - 2a^2 y_0^3 - 10ia y_0^2 + 3a z_0 + y_0, \\ P_3 &= 2a^3 y_0^3 z_0 + 6ia^2 y_0^2 z_0 + 2a^2 w_0 y_0 + 4a^2 y_0^4 \\ &\quad + 2ia w_0 - 4ia y_0^3 - 2a y_0 z_0 + 2iz_0 + 7y_0^2, \\ P_4 &= 2a^3 y_0^4 z_0 + 8ia^2 y_0^3 z_0 + a^2 w_0 y_0^2 + 4a^2 y_0^5 \\ &\quad + 2ia w_0 y_0 + 7ia y_0^4 - 10a y_0^2 z_0 - 4iy_0 z_0 - w_0 - 3y_0^3, \\ P(a, y_0, z_0, w_0) &= (P_0, P_2, P_3, P_4). \end{aligned} \quad (\text{EZ4})$$

For a target $u = (u_0, u_2, u_3, u_4)$, write

$$H_u(U, V) = u_0U^4 + U^3V + u_2U^2V^2 + u_3UV^3 + u_4V^4, \quad h_u(T) = H_u(T, 1). \quad (\text{EZ5})$$

The full inverse, image complement and nonproper-value locus of EZ4 are proved in the cumulative source [3, GF1–GF18]. We state every inverse formula used below, so the crosswalk does not depend on an unnamed analogy.

2 The normalized prime–denominator quartic

Definition 2.1 (Arithmetic quartic). Define

$$\mathcal{J}(p; x, y, z) = -\frac{1}{S}(U - pV)(U - xV)(U - yV)(U - zV). \quad (\text{EZ6})$$

The minus sign and the factor S^{-1} are retained: they make the coefficient of U^3V exactly one, as required by EZ5.

Theorem 2.2 (Exact coefficients and intrinsic prime). *The coefficient target $J_4(p; x, y, z) = (u_0, u_2, u_3, u_4)$ of EZ6 is*

$$\boxed{u_0 = -\frac{1}{S}, \quad u_2 = -\frac{ps_1 + s_2}{S}, \quad u_3 = \frac{ps_2 + s_3}{S} = \frac{5s_3}{S}, \quad u_4 = -\frac{ps_3}{S}.} \quad (\text{EZ7})$$

In particular $u_0u_3u_4 \neq 0$, and the prime survives the passage to the unmarked coefficient target:

$$\boxed{p = -\frac{5u_4}{u_3}.} \quad (\text{EZ8})$$

Proof. Expanding the product in EZ6 gives

$$-S^{-1}(U^4 - SU^3V + (ps_1 + s_2)U^2V^2 - (ps_2 + s_3)UV^3 + ps_3V^4).$$

This proves the first expressions in EZ7. Relation EZ3 gives $ps_2 + s_3 = 5s_3$. All source integers are positive, so the displayed denominators and numerators do not vanish. Dividing the last two coefficient identities gives EZ8. $\square$

Theorem 2.3 (Coefficient hypersurface and exact converse). *On the open target chart $D(u_0u_3u_4)$, every arithmetic quartic satisfies*

$$\boxed{G(u) = 625u_0u_4^3 - 125u_3u_4^2 + 25u_2u_3^2u_4 - 4u_3^4 = 0.} \quad (\text{EZ9})$$

Conversely, if $u_0u_3u_4 \neq 0$ and EZ9 holds, then $p = -5u_4/u_3$ is a root of h_u . If the other roots in an algebraic closure are denoted x, y, z , counted with multiplicity, then

$$p(xy + xz + yz) = 4xyz. \quad (\text{EZ10})$$

When all four roots are nonzero, EZ10 is equivalent to EZ1.

Proof. Direct substitution of $p = -5u_4/u_3$ into EZ5 gives

$$h_u(p) = \frac{u_4}{u_3^4}G(u).$$

The prefactor is nonzero on the stated chart, proving both directions of the root assertion. Factor $h_u(T) = u_0(T - p)(T - x)(T - y)(T - z)$. Vieta's formulas give $u_3 = -u_0(ps_2 + s_3)$ and $u_4 = u_0ps_3$, where now $s_2 = xy + xz + yz$, $s_3 = xyz$. Equation EZ8, which follows here from the definition of p , becomes $p = 5ps_3/(ps_2 + s_3)$. Since $p \neq 0$, this is exactly EZ10. Dividing EZ10 by $pxyz$ proves the last assertion. $\square$

Remark 2.4 (What the hypersurface does not encode). The equation $G = 0$ is the Zariski equation of the normalized reciprocal relation on this open chart. It does not force the four roots to be positive integers, make p prime, select an ordering of x, y, z , or impose any exterior, middle, congruence or divisibility gate. These arithmetic conditions remain part of the source. No admission criterion is being invented for the published Fable theorem; this is the exact image statement for the new ES map.

3 The eight states and the prime-marked pair

Let $t_1 = p, t_2 = x, t_3 = y, t_4 = z$, and for each root put

$$\Delta_j = \prod_{k \neq j} (t_j - t_k). \quad (1)$$

Theorem 3.1 (Complete fibre). *If p, x, y, z are pairwise distinct, then J_4 lies in the finite étale domain*

$$\mathcal{B} = \{u : u_0 \text{Disc}(H_u) \neq 0\} \quad (\text{EZ11})$$

of EZ4, and $P^{-1}(J_4)$ consists of exactly eight points. For every $j \in \{1, 2, 3, 4\}$ and each of the two signs, choose

$$a_{j,\sigma}^2 = -\frac{S}{\Delta_j}, \quad a_{j,-} = -a_{j,+}. \quad (\text{EZ12})$$

The corresponding source point is

$$\begin{aligned} q_{j,\sigma} &= (a_{j,\sigma}, y_{j,\sigma}, z_{j,\sigma}, w_{j,\sigma}), \\ y_{j,\sigma} &= -t_j - \frac{i}{a_{j,\sigma}}, \\ z_{j,\sigma} &= \frac{u_0}{a_{j,\sigma}^3} + \frac{2t_j}{a_{j,\sigma}} + \frac{3i}{a_{j,\sigma}^2}, \\ w_{j,\sigma} &= \frac{7it_j^2}{a_{j,\sigma}} + \frac{u_2 - 17t_j + u_0t_j^2}{a_{j,\sigma}^2} - \frac{13i}{a_{j,\sigma}^3} - \frac{2u_0}{a_{j,\sigma}^4}. \end{aligned} \quad (\text{EZ13})$$

These eight points are distinct and exhaust the fibre.

Proof. The leading coefficient is $u_0 = -S^{-1} \neq 0$, and

$$\text{Disc}(H_{J_4}) = S^{-6} \prod_{1 \leq j < k \leq 4} (t_j - t_k)^2. \quad (\text{EZ14})$$

Thus the distinctness hypothesis is exactly EZ11. At the root t_j ,

$$h'_{J_4}(t_j) = -S^{-1}\Delta_j,$$

so the inverse equation $a^2h'(t_j) = 1$ is precisely EZ12. Comparing coefficients in a factorization $H = LC$, where $L = a(U - t_jV)$, gives EZ13. Substitution into EZ4 returns J_4 . Conversely, the complete inverse theorem for EZ4 says that on $u_0 \neq 0$ every source point selects a simple root $t = -b/a$ and a sign satisfying $a^2h'(t) = 1$. Four simple roots and two signs therefore give exactly eight points, with no $a = 0$ chart. $\square$

Corollary 3.2 (The two prime-marked lifts). *Put*

$$\Delta_p = (p - x)(p - y)(p - z), \quad \lambda^2 = -\frac{S}{\Delta_p}, \quad \mu = -\frac{1}{\lambda S}. \quad (\text{EZ15})$$

Then

$$L = \lambda(U - pV), \quad C = \mu(U - xV)(U - yV)(U - zV) \quad (\text{EZ16})$$

satisfy

$$LC = \mathcal{J}(p; x, y, z), \quad \text{Res}(L, C) = 1, \text{quad}ad + bc = 1, \text{quad}4af - be = 0. \quad (\text{EZ17})$$

The two choices of λ are exactly $q_{p,+}$, $q_{p,-}$, and the rational deck involution exchanges them. The other six fibre points mark x, y, z ; they are not additional prime-marked ES inverses.

Proof. The product identity follows from $\lambda\mu = -S^{-1}$. In the coefficient convention $L = aU + bV$, $C = cU^3 + dU^2V + eUV^2 + fV^3$, EZ16 gives

$$(a, b, c, d, e, f) = (\lambda, -\lambda p, \mu, -\mu s_1, \mu s_2, -\mu s_3).$$

The retained resultant orientation is

$$\text{Res}(L, C) = C(-b, a) = \lambda^3 \mu \Delta_p = -\lambda^2 \Delta_p / S = 1.$$

Furthermore $ad + bc = -\lambda\mu(s_1 + p) = 1$, and $4af - be = \lambda\mu(ps_2 - 4s_3) = 0$ by EZ3. The source formula in EZ13 is the inverse of these six factor coordinates. The deck statement is GF18 of [3]. $\square$

4 The complete Turn 7 exterior map

The exterior source retained in the Turn 7 work has

$$a = hrs, \text{quad}u = hr^2, \text{quad}R = 4a - p, \text{quad}D = \frac{4u + 1}{R} = \frac{r + \kappa}{s}, \text{quad}R\kappa = pr + s, \quad (\text{EZ18})$$

and ordered denominators

$$(x, y, z) = (hrs, hsk, phr\kappa). \quad (\text{EZ19})$$

Theorem 4.1 (Exterior states land in the eight-sheet domain). *Every actual Turn 7 exterior state maps by EZ19 and EZ6 into $\mathcal{B}$. More precisely,*

$$x < p, \text{qquad}x < y < z, \text{qquad}z > p, \text{qquad}y \neq p, \quad (\text{EZ20})$$

so p, x, y, z are pairwise distinct and EZ14 is strictly positive.

Proof. The exterior inequalities give $p/4 < a = x < p/2$, hence $x < p$. The retained normalization gives $r < \kappa$, so $y/x = \kappa/r > 1$. Since $s = a/g \leq a < p/2$, while $pr \geq p$, we have $z/y = pr/s > 1$. Also $z/p = hr\kappa > 1$. Finally $p \nmid h$ and $p \nmid s$, because both divide the positive integer $a < p$. If $p \mid \kappa$, then $R\kappa = pr + s$ would imply $p \mid s$, a contradiction. Therefore $p \nmid hsk = y$, so $y \neq p$. This proves EZ20 and the claim. $\square$

The marked shape curve retains

$$X = v, \text{quad}Y = 2a, \text{quad}R = X - \alpha, \text{quad}D = X - \beta, \text{quad}p = 2Y - R, \text{quad}u = (RD - 1)/4. \quad (\text{EZ21})$$

On its stated integral image, the inverse

$$g = \gcd(a, u), \quad (h, r, s) = (g^2/u, u/g, a/g), \quad \kappa = (pr + s)/R \quad (\text{EZ22})$$

followed by EZ19 and EZ6 is the complete typed map from the Turn 7 marked cubic to J_4 . Algebraically, the coefficient target forgets the action of S_3 permuting x, y, z . On the actual strict real chamber EZ20, sorting the three non-prime roots restores $x < y < z$. One then reconstructs

$$R = 4x - p, \text{quad}u = \frac{xz}{py} = \frac{x^2}{Ry - px}, \text{quad}D = \frac{4u + 1}{R}, \text{quad}v = \frac{x^2}{u}, \text{quad}(X, Y) = (v, 2x), \quad (\text{EZ23})$$

and EZ22 restores every exterior marking. Outside the arithmetic chamber, there is no asserted global rational inverse from an unmarked root set to the ordered integral source.

5 The reciprocal suspension is a different exact bridge

For comparison, an ordered ES solution also gives reciprocal roots

$$t_1 = p/x, \text{quad}t_2 = p/y, \text{quad}t_3 = p/z, \text{quad}t_1 + t_2 + t_3 = 4, \text{quade}_2 = \sum_{i < j} t_i t_j, \text{quade}_3 = t_1 t_2 t_3. \quad (\text{EZ24})$$

The normalized cubic

$$C_{\text{ES}}(U, V) = -\frac{1}{4} \prod_{j=1}^3 (U - t_j V)$$

has the canonical suspension

$$\mathfrak{S}(C_{\text{ES}}) = -4VC_{\text{ES}} = V \prod_{j=1}^3 (U - t_j V), \text{quad}J_3 = (0, -4, e_2, -e_3). \quad (\text{EZ25})$$

If the three roots are distinct, the fibre has exactly seven points: two over each finite root and the single infinity-chart point

$$q_\infty = \left(0, -4, \frac{i(112 - e_2)}{2}, -704 + 8e_2 + e_3\right). \quad (\text{EZ26})$$

For $\Delta_j = \prod_{k \neq j} (t_j - t_k)$, the six finite points are EZ13 with $u_0 = 0, u_2 = -4, t_j$ as root and $a^2 = \Delta_j^{-1}$. Its discriminant is

$$\prod_{i < j} (t_i - t_j)^2 = \frac{p^6 (x - y)^2 (x - z)^2 (y - z)^2}{x^4 y^4 z^4}. \quad (\text{EZ27})$$

Unlike J_4 , every J_3 lies on the $u_0 = 0$ component of the exact nonproper locus

$$\mathcal{N}(P) = \{u : u_0 \text{Disc}(H_u) = 0\}. \quad (\text{EZ28})$$

Thus the eight-state quartic and seven-state suspension are related maps with different retained data, fibres and exceptional loci. Neither is substituted for the other.

6 The exact bridge to a fixed weighted conductor

Fix one reference arithmetic quartet with distinct roots $\rho_{*,1}, \dots, \rho_{*,4}$, its original exponential polynomial, first nonzero moment μ_v , period frame and Taylor unit. Let

$$V_* = (\rho_{*,j}^n)_{0 \leq n \leq 3, 1 \leq j \leq 4}, \quad (B_{v,*})_{kj} = \begin{cases} \binom{v+j}{k} \mu_{v+j-k}, & 0 \leq k \leq j \leq 3, \\ 0, & k > j, \end{cases} \quad (\text{EZ29})$$

and retain the source frame

$$K = \begin{pmatrix} 0 & i & 1/\sqrt{2} & 1/\sqrt{2} \\ 0 & -1 & -1 - \sqrt{2}i & 1 - \sqrt{2}i \\ i/2 & -3i & 2\sqrt{2} + 6i & -2\sqrt{2} + 6i \\ 0 & 13 & -34 - 19\sqrt{2}i & 34 - 19\sqrt{2}i \end{pmatrix}. \quad (\text{EZ30})$$

Define

$$\Phi_* = (V_*^\top)^{-1} K^{-1}, \quad \Psi_* = B_{v,*} (V_*^\top)^{-1} K^{-1}. \quad (\text{EZ31})$$

The columns of K are the auxiliary labelled source frame; they are not identified with either p, x, y, z or the reference roots. The original weighted conductor is $T_{A,*}|_{\mathcal{U}_*}$, and the exact square is

$$T_{A,*} \Phi_* = \Psi_*. \quad (\text{EZ32})$$

The receiving map Ψ_* is not itself the conductor.

Theorem 6.1 (Conductor transport of the ES quartic). *All three maps in EZ32 are invertible and have zero kernel. Put*

$$P_{\mathcal{U},*} = \Phi_* P \Phi_*^{-1}, \quad P_{\mathcal{W},*} = \Psi_* P \Psi_*^{-1}. \quad (\text{EZ33})$$

For every target u and source point q with $P(q) = u$,

$$P_{\mathcal{U},*}(\Phi_* q) = \Phi_* u, \quad q \text{quad} P_{\mathcal{W},*}(\Psi_* q) = \Psi_* u, \quad q \text{quad} T_{A,*} P_{\mathcal{U},*} = P_{\mathcal{W},*} T_{A,*}. \quad (\text{EZ34})$$

In particular the eight points EZ13 above J_4 , their deck involution, and the exceptional set EZ28 pass bijectively to the fixed conductor planes.

Proof. Distinct reference roots make V_* invertible. Moreover

$$\det B_{v,*} = \mu_v^4 \prod_{j=0}^3 \binom{v+j}{v} \neq 0, \quad \det K = 77\sqrt{2}i/2.$$

Thus EZ31 is invertible and EZ32 is the exact conductor identity proved in [3, FC26–FC35]. The first two identities in EZ34 are the definitions of conjugacy. Inserting EZ32 between them proves the last one. Invertible linear maps preserve fibres, escaping sequences, convergence and deck equations, proving the final assertion. $\square$

For fixed $s > 0$ and centre c_* , the receiving plane retains the norm

$$\|q\|_{\Gamma,*}^2 = \int_{\mathbb{R}} |q(c_* + iy)|^2 \frac{(2\pi)^{s/2} 2^{s/2}}{4\pi\Gamma(s/2)} |\Gamma(s/4 + iy/2)|^2 dy. \quad (\text{EZ35})$$

Its Meixner–Pollaczek weight, mass, recurrence and generating function are the exact specialization of DLMF 18.19.8–9, 18.22.8 and 18.23.7 [4]. This metric makes the bridge quantitative; it does not make the four ES roots into zeta zeros.

7 A root-algebra bridge and its exact projective obstruction

After choosing a bijection $t_j \leftrightarrow \rho_{*,j}$ between the four distinct ES roots and the four fixed reference roots, let $\ell_j^\rho(S)$ be the Lagrange polynomial satisfying $\ell_j^\rho(\rho_{*,k}) = \delta_{jk}$. Then

$$\Theta : \frac{\mathbb{C}[T]}{\prod_j (T - t_j)} \longrightarrow \frac{\mathbb{C}[S]}{\prod_j (S - \rho_{*,j})}, \quad [f] \longmapsto \sum_{j=1}^4 f(t_j) [\ell_j^\rho(S)] \quad (\text{EZ36})$$

is a unital algebra isomorphism. Its inverse evaluates at the reference roots and interpolates at the ES roots; hence both kernels are zero. It depends on the chosen bijection. It is generally not induced by one fractional-linear map. Such a map exists exactly when the corresponding ordered cross ratios agree, up to the six values produced by relabelling. This is the exact projective obstruction; failure of that stronger map does not erase EZ31–36.

8 Scope

The new composite is

$$\begin{aligned} & \text{actual ordered ES exterior state} \\ & \quad \downarrow \text{EZ18–23} \\ & \quad (p; x, y, z) \\ & \quad \downarrow \text{normalized product EZ6} \\ & J_4 \in \mathcal{B}, \quad p = -5u_4/u_3, \quad G(J_4) = 0 \\ & \quad \downarrow P^{-1} \\ & \quad \{q_{p,\pm}, q_{x,\pm}, q_{y,\pm}, q_{z,\pm}\} \\ & \quad \downarrow (\Phi_*, \Psi_*, T_{A,*}) \\ & \text{eight states in the fixed weighted-conductor square.} \end{aligned}$$

The map begins with an actual ES witness and therefore is not a proof that a witness exists for every p . The hypersurface $G = 0$ is not by itself an arithmetic occupancy theorem. The fixed quartet and fixed conductor data do not vary with p, x, y, z , and the construction proves no zero of the Riemann zeta function. The signed-root monodromy group of order 192 belongs to the generic eight-sheet cover over $\mathcal{B}$; it is not attributed to the seven-state boundary suspension.

References

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